

Signal Generators

Many engineers believe that all their design and testing requirements are met just by acquisition instruments like oscilloscopes and logic analysers. But a measurement setup is complete only when it has both acquisition and stimulus systems to complement each other. A signal generator pairs with the acquisition system to produce the most reliable test and measurement results



(Image courtesy: www.ece.lsu.edu)

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A signal generator produces ideal waveforms that act as a stimulus for electronic measurement. Most circuits require some kind of signal to be fed at the input to study and test their output using an acquisition system comprising various instruments like oscilloscope, multimeter and logic analyser. The output makes sense only when you have an ideal and controlled input.

Signal generators can provide a variety of signals for testing, which can be controlled as per your requirement. The signal can be a sine wave, a digital pulse, a binary pattern, an analogue function or any other arbitrary waveform.

One of the greatest advantages of a signal generator is that it can even add a controlled distortion to the signal. This is highly useful to study the

response of your circuit in presence of the distorted signal, as you can check how much your design can handle and accordingly decide the correct performances range (refer Fig. 1).

Why do you need one?

An acquisition system together with a stimulus system like signal generator makes up a complete testing solution that can feed a device with real-world signals and acquire and analyse the resulting outputs. You can measure

the output at various test points easily but it makes sense only when you can control what goes in, and that you can only do when a signal generator is connected at the input.

Signal generators find various applications, all of which fall under three main categories: margin testing, verification and characterisation.

Margin testing. Engineers need to test their designs in order to ensure that these meet the design specifications across the full range of operation. This is known as margin testing. This is a design challenge that requires a complete setup with stimulus and acquisition systems. With such a system you vary all the input parameters using the stimulus and check the limits until which you get the acceptable output. This testing also identifies how much stress the device can take while still doing its reliable operation.

Characterisation. Every device comes with characterisation information provided by its manufacturer. Characterisation information is actually a behaviour summary of the device. This information becomes available after the device has undergone complete characterisation tests and the data has been analysed. The characterisation

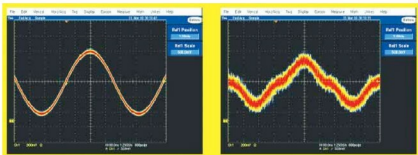


Fig. 1: Ideal vs waveform after adding controlled distortion